

Jiyu Hu

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Education

University of Illinois Urbana-Champaign <i>Ph.D. in Computer Science</i>	Aug. 2023 – Present Urbana, IL
Carnegie Mellon University – School of Computer Science <i>Master of Computational Data Science</i>	Aug. 2021 – May 2023 Pittsburgh, PA
University of Illinois Urbana-Champaign <i>Bachelor of Science in Computer Engineering</i>	Aug. 2017 – May 2021 Urbana, IL

Publications

* = Equal Contribution

Jiyu Hu, Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan. MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory. In *20th USENIX Symposium on Operating Systems Design and Implementation, OSDI '26*. USENIX Association, 2026. (To Appear)

Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, and Aishwarya Ganesan. LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications. *ACM Transactions on Computer Systems*, August 2025

Shreesha G. Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, and Ramnatthan Alagappan. Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering. In *19th USENIX Symposium on Operating Systems Design and Implementation, OSDI '25*, pages 465–481. USENIX Association, 2025

Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, and Aishwarya Ganesan. LazyLog: A New Shared Log Abstraction for Low-Latency Applications. In *Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles, SOSP '24*, page 296–312, New York, NY, USA, 2024. Association for Computing Machinery

Jiyu Hu, Jack Kosaian, and K. V. Rashmi. Rethinking Erasure-Coding Libraries in the Age of Optimized Machine Learning. In *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems, HotStorage '24*, page 23–30, New York, NY, USA, 2024. Association for Computing Machinery

Rui Yang, Jiangran Wang, Jiyu Hu, Shichu Zhu, Yifei Li, and Indranil Gupta. Medley: A Membership Service for IoT Networks. *IEEE Transactions on Network and Service Management*, 19(3):2492–2505, 2022

Xueda Shen*, Jiyu Hu*, Yunqi Zhang, and Ian C. Quinn. B2-Coupon: Efficient and Non-intrusive Mobile Coupon Distribution using Dual Bloom Filter. In *2020 IEEE/ACM Symposium on Edge Computing (SEC)*, pages 358–363, 2020

Presentations

Rethinking Erasure Coding Libraries in the Age of Optimized Machine Learning <i>HotStorage '24</i>	Jul. 2024 Santa Clara, CA
B² Coupon <i>ACM SEC (workshop talk)</i>	Nov. 2020

Research Experience

- MEGALON** Apr. 2024 – Dec. 2025
DASSL, University of Illinois Urbana-Champaign Urbana, IL
- Investigate the design of efficient sharing of CXL memory under the assumption of partial hardware coherence.
 - Design a software cache coherence mechanism utilizing shared-log-enabled replicated metadata to support a large number of shared objects.
- SpecLog** Apr. 2024 – Dec. 2024
DASSL, University of Illinois Urbana-Champaign Urbana, IL
- Design and implement a new shared log abstraction that significantly decreases e2e latency by speculation.
- LazyLog** Jan. 2024 – Jul. 2024
DASSL, University of Illinois Urbana-Champaign Urbana, IL
- Design and implement a new shared log abstraction that significantly decreases append latency from state-of-the-art shared log implementations by delaying ordering the appended log entries.
- TVM-EC** Jan. 2022 – Jul. 2024
TheSys Lab, Carnegie Mellon University Pittsburgh, PA
- Propose a new way of implementing high-performance erasure-coding libraries via machine learning libraries, reducing the effort to design and maintain erasure coding libraries.
- Medley** Jan. 2020 – Jul. 2021
Distributed Protocols Research Group, University of Illinois Urbana-Champaign Urbana, IL
- Develop and evaluate a new IoT failure detection protocol that is aware of the spatial locality of physical nodes so as to decrease the overall communication overhead in an unstable network environment.
- B²-Coupon** Nov. 2019 – Nov. 2020
Prof. Dong Xuan's Research Group, The Ohio State University Columbus, OH
- Design a better coupon distribution protocol for mobile devices.

Awards & Scholarships

- Student Travel Grant**
OSDI '25 2025
- Best Paper Award (LazyLog)**
SOSP '24 2024
- Senior Design Instructor's Award (The Best Senior Design Award)**
University of Illinois Urbana-Champaign 2021
- Bradley A. Simons Memorial Scholarship**
University of Illinois Urbana-Champaign 2019
- Dean's List**
University of Illinois Urbana-Champaign 2017, 2018, 2019, 2020, 2021

Services

- CAIS '26, Artifact Evaluation Committee
- EuroSys '26, Shadow PC
- SOSP '25, Artifact Evaluation Committee
- FAST '26, Artifact Evaluation Committee
- OSDI '24, ATC '24, Artifact Evaluation Committee

Teaching

CS 498 Cloud Computing Applications
University of Illinois Urbana-Champaign

TA
2026

Industry Experience

GSL, Microsoft Research

May 2026 – Aug. 2026

Research Intern (Supervisor: Rathijit Sen)

- To work on disaggregated memory.

USM, Exadata, Oracle America, Inc.

May 2022 – Aug. 2022

Software Intern

- Worked on the kernel log aggregation and analysis framework of Oracle distributed database.

CUDA Math Library, NVIDIA Corporation

June 2020 – Aug. 2020

Software Intern

- Analyzed the floating point error propagation in cuBLAS GEMM kernel.

Skills

Languages: C, C++, Rust, x86, Go, Java, Python

Tools & frameworks: RDMA, CXL, Git, Docker, Kubernetes, CUDA